EXERCISE 9

Hyperbolic Motion • Variable Hanging Mass

Falling Chain from Table and Hyperbolic Motion

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PHASE 1: STATEMENT & SETUP

> A chain of length $\ell = 5$ m and mass m, with uniform mass density, is sitting on a table. If the chain is initially at rest and 1 m of the chain hangs from the edge of the table, apply Newton's second law to calculate the time t_* when the whole chain falls off the table, neglecting any friction.
>
> *Hint:* Formulate a second order ODE for the variable $x(t)$, which is the length of the portion of chain that is overhanging, as a function of time.

PHASE 2: FUNDAMENTAL FORMULATION

Let $x(t) \in [1, 5]$ be the overhanging length. Mass density is $\lambda = m/\ell$. The downward gravity force pulling the chain is $F_{\text{ext}} = m_{\text{hang}} g = (\lambda x) g = \frac{m}{\ell} g x$. Because the table is frictionless and the entire chain of constant total mass m accelerates together along its contour:

$$m\ddot{x} = \frac{m}{\ell} g x \implies \ddot{x} - \frac{g}{\ell} x = 0$$

PHASE 3: STEP-BY-STEP ANALYTICAL DERIVATION

Step 1: Auxiliary Equation and General Solution

Let $\alpha = \sqrt{\frac{g}{\ell}}$. The ODE is $\ddot{x} - \alpha^2 x = 0$.

Auxiliary eq: $r^2 - \alpha^2 = 0 \implies r = \pm\alpha$.

Hyperbolic general solution:

$$x(t) = c_1 \cosh(\alpha t) + c_2 \sinh(\alpha t)$$

$$\dot{x}(t) = \alpha\left[c_1 \sinh(\alpha t) + c_2 \cosh(\alpha t)\right]$$

Step 2: Initial Conditions

Released from rest: $\dot{x}(0) = \alpha c_2 = 0 \implies c_2 = 0$.

Initial hanging length: $x(0) = c_1 = x_0 = 1\ \text{m}$.

$$\mathbf{x(t) = x_0 \cosh(\alpha t) = \cosh\left(\sqrt{\frac{g}{\ell}}\, t\right)}$$

Step 3: Detachment Time t_*

The entire chain falls off when $x(t_*) = \ell = 5\ \text{m}$:

$$\cosh(\alpha t_*) = \frac{\ell}{x_0} = \frac{5}{1} = 5$$

$$\alpha t_* = \operatorname{arcosh}(5) = \ln\left(5 + \sqrt{5^2 - 1}\right) = \ln(5 + \sqrt{24}) = \ln(5 + 2\sqrt{6})$$

Step 4: Numerical Evaluation

Using standard gravity $g = 9.8\ \text{m/s}^2$ and $\ell = 5\ \text{m}$:

$$\alpha = \sqrt{\frac{9.8}{5}} = \sqrt{1.96} = 1.4\ \text{s}^{-1}$$

$$\ln(5 + 2\sqrt{6}) \approx \ln(5 + 4.898979) = \ln(9.898979) \approx 2.292432$$

$$\mathbf{t_* = \frac{2.292432}{1.4} \approx 1.63745\ \text{s} \approx 1.637\ \text{s}}$$

(If taking $g = 9.81\ \text{m/s}^2$, $\alpha \approx 1.40071\ \text{s}^{-1} \implies t_* \approx 1.6366\ \text{s} \approx 1.637\ \text{s}$)

PHASE 4: INTERPRETATION & CONSERVATION CHECK

MASTER FALL TIME & DETACHMENT VELOCITY

$$\boxed{t_* = \frac{\operatorname{arcosh}(5)}{1.4} \approx 1.637\ \text{s}}$$

Detachment velocity: $v_* = \dot{x}(t_*) = x_0 \alpha \sinh(\alpha t_*) = 1 \times 1.4 \times \sqrt{5^2 - 1} = 1.4\sqrt{24} \approx \mathbf{6.859}\ \text{m/s}$.

Energy Check: $\Delta E_p = mg\frac{\ell}{2} - mg\frac{x_0^2}{2\ell} = \frac{mg}{2\ell}(\ell^2 - x_0^2) = \frac{1}{2}mv_*^2 \implies v_* = \sqrt{\frac{g}{\ell}(\ell^2 - x_0^2)} = \sqrt{1.96(24)} \approx 6.859\ \text{m/s}$. Exact match!